

FLEX-DBS manuscript: working notes

Working notes

Data still needed

- **Electrode construction (Section 3.1).** Materials, wire diameter, insulation, exposed length or tip geometry, bilateral spacing, return electrode, connector and fabrication steps. Planned figure: electrode photograph and schematic.
- **Surgery, animals and ethics.** Ethics statement, strain, sex, age, housing, stereotaxic target and coordinates, anesthesia and analgesia, electrode exposed surface area, IACUC protocol number, and the implantation date that anchors “weeks after implantation” in Section 3.
- **In-vivo validation (Sections 6.1–6.3).**
- **Abstract and Conclusions.** Update the Results and Conclusions once in-vivo data exist; both currently describe design, 1 k Ω bench, impedance and battery results only.

Impedance measurement

- Data come from the platinum cohort 1 (M1, M2) and platinum cohort 2 (F1, F2) sheets. Excluded: M1 output 0 (near 0 V throughout, likely a short) and both electrodes of F1 (unstable; 47–118 k Ω and 23–106 k Ω between sessions, with within-session drift at 7 weeks). Analyzed: five electrodes from M1, M2 and F2. Confirm the wording of the exclusion criteria, and consider stating a quantitative criterion.
- The StimJim bench check (2/10/20/68 k Ω) showed its voltage read-back 6–73 % below the oscilloscope, which biases the in-vivo values low, while the 450 μ s pulse biases them high. Consider remeasuring with an impedance meter (for example, a NanoZ at 1 kHz) or with an oscilloscope on the StimJim output, and report both.
- The 450 μ s measurement pulse is longer than the DBS pulses; a measurement at the stimulation pulse width would better match the compliance calculation.
- The two-electrode configuration is justified with Koo and Weiland (2018); confirm that this is the framing the authors want.

Citations to add or verify

- Paulat et al. (2026) is still a Research Square preprint (posted 2026-04-27; checked 2026-10-01). Update if it is published.

Table 4

- Check whether the Pinnell et al. (2018) Figshare release (described as design files and assembly instructions) contains firmware source.
- Supplementary files were not read for Burton et al. (2021) or Shen et al. (2026), and Paulat et al. (2026)’s programmable ranges and battery models come from its supplement.

Firmware and design

- Firmware v0.5 delivers the 130 Hz setting as 142.5 Hz, and at 180 μ s the stimulating phase measures ~ 60 μ s long and the recovery phase 22–46 μ s short, leaving a 26–35 % net charge per pulse (Section 4.1). If the fixes listed in Section 7 are made before posting, remeasure and update Section 4.1, Figure 5, Table 2 and the Limitations.
- The photograph in Figure 2A shows the “v8.1” board silkscreen; decide whether to mention the hardware revision in the text.
- Supplementary S1: consider adding the BOM (v8.1/BOM.csv) alongside the schematic.

End matter

- Confirm author order, middle initials and affiliations.
- Funding statement is pending.
- Data availability: decide whether to deposit the impedance data and figure scripts in a public repository instead of “available on request”.
- Before journal submission (not needed for bioRxiv): Highlights, graphical abstract, CRediT statement, generative-AI declaration, line numbers and double spacing per the Journal of Neuroscience Methods guide.

References

- Burton, A., Won, S.M., Sohrabi, A.K., Stuart, T., Amirhossein, A., Kim, J.U., Park, Y., Gabros, A., Rogers, J.A., Vitale, F., Richardson, A.G., Gutruf, P., 2021. Wireless, battery-free, and fully implantable electrical neurostimulation in freely moving rodents. *Microsystems & Nanoengineering* 7, 62. doi:10.1038/s41378-021-00294-7.
- Koo, B., Weiland, J., 2018. Experimental factors effecting stability of electrochemical impedance spectroscopy measurements, in: 2018 40th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC), pp. 2949–2952. doi:10.1109/EMBC.2018.8512973.
- Paulat, R., Koschützke, L., Haumesser, J., Maurer, L., Altschüler, J., Tang, H., Hounchonou, H., Daugardt, A., Harnack, D., Just, V., Schulz, G., Harms, C., Kühn, A.A., Wenger, N., van Riesen, C., 2026. Wireless lightweight implantable pulse generators enable chronic dual-target deep brain stimulation in rodent disease models. *Research Square preprint*. doi:10.21203/rs.3.rs-9019884/v1.
- Pinnell, R.C., Pereira de Vasconcelos, A., Cassel, J.C., Hofmann, U.G., 2018. A miniaturized, programmable deep-brain stimulator for group-housing and water maze use. *Frontiers in Neuroscience* 12, 231. doi:10.3389/fnins.2018.00231.
- Shen, Y., Zhao, D., Lv, Y., Liu, Y., Chen, Y., Zhang, Y., Chen, S., 2026. A wearable deep brain stimulation system for behavioral studies in rodents. *Frontiers in Neuroscience* 19, 1707636. doi:10.3389/fnins.2025.1707636.